Approach Document: Predicting Ride Completion in NCR Bookings

Project Title

Predicting Ride Completion in NCR Bookings: Optimizing Dispatch and Reducing Operational Risk

Goal & Context

The primary goal is to build a high-accuracy predictive model to estimate the probability that a ride request, once accepted, will be successfully **Completed**. This prediction is vital for reducing **driver churn** (by minimizing fruitless pickups), managing **customer expectation** (by reducing estimated time of arrival (ETA) uncertainty), and proactively implementing **dynamic interventions** (e.g., penalty adjustments, pre-emptive re-dispatch) to improve the platform's reliability.

Dataset & Task

The dataset, ncr_ride_bookings.csv (150,000 records), contains detailed ride request, customer, and outcome information.

Feature Type	Key Features	Task
Temporal	Date, Time	Feature engineering for peak hours, day of week, seasonality.
Geographic	Pickup Location, Drop Location	Feature engineering for high-demand zones, low-supply areas.
Historical	Cancelled Rides by Customer, Cancelled Rides by Driver	Feature engineering for individual behavioral risk scoring.
Outcome	Booking Status	Binary Classification Target (Y): Completed vs. Non-Completed (Cancelled, No Driver Found, Incomplete).

Section A: Business Understanding & Data Audit

A1. Business Problem Definition & KPIs

The core business problem is **uncertainty in service delivery**. A high non-completion rate leads to lost revenue, increased operational costs (penalty reimbursements), and poor customer/driver satisfaction.

The relevant Key Performance Indicators (KPIs) are:

- **Model Performance: Area Under the Curve (AUC)** (robust measure of discriminative power), **F1-Score** (to balance Precision and Recall for a balanced outcome).
- **Business Impact: Cost of Misclassification** (monetary value lost due to False Negatives/False Positives) and **Reduction in Predicted Non-Completion Rate**.

A2. Identify Target Variable & Leakage

- **Target Variable:** Booking Status binarized into: 1 (Completed) and 0 (Non-Completed).
- **Input Features:** All columns available at the moment of booking/driver acceptance.
- **Data Leakage Risk (Critical):** Any feature whose value is only known *after* the ride outcome is determined represents leakage.
    - **High Leakage Risk:** Driver Ratings, Customer Rating, Booking Value, Ride Distance, and Payment Method (only available if the ride proceeds/completes) must be **treated carefully** or excluded from the model to predict the *initial* acceptance/cancellation risk.

A3. Perform Data Audit

The dataset consists of 150,000 rows. A key observation is the presence of null values that are **outcome-dependent**.

- Columns like Booking Value, Ride Distance, and Avg CTAT have only 102,000 non-null values, strongly suggesting they are primarily recorded for completed or semi-completed rides.
- The non-completion reason columns (Reason for cancelling by Customer, etc.) are sparsely populated, which is expected as they are only relevant for non-completed bookings.

A4. Describe Five Important Variables (Derived)

Derived Variable	Impact on Prediction & Business Decision
Time Slot of Day	Temporal Feature: Binned from Time (e.g., Morning Peak, Midday, Evening Rush). This predicts demand/supply stress, which directly affects completion probability due to driver availability.

Pickup Location Density Score	Geographical Feature: Calculated based on the volume of bookings/cancellations in a zone. High density areas may have higher driver supply, increasing completion chance.
Customer Cancellation Rate (Historical)	Behavioral Feature: Derived from Customer ID and historical cancellation columns. This is the single strongest indicator of future cancellation risk.
Trip Duration (Predicted)	Monetary Feature: Derived from Ride Distance (if available pre-completion) and average speed. Longer/more valuable trips may have higher driver retention but higher risk of passenger cancellation.
Day Type	Temporal Feature: Categorized from Date (e.g., Weekend, Public Holiday, Normal Weekday). Directly informs the likelihood of traffic or unusual demand patterns.

Section B: Data Cleaning & Preprocessing

B1. Handle Missing Values (Focus on Leakage)

The imputation strategy must address the **data leakage** identified in A2.

- **Outcome-Dependent Columns (Leakage Risk):** For features like Booking Value or Ride Distance, if the ride is marked "No Driver Found," the value is legitimately missing. Simply imputing the mean for all rows would introduce bias. **Strategy:** A new feature, Is_Booking_Value_Available (a binary flag), should be created, and missing values in the original column should be imputed with a value (like -1 or 0) that signals missingness *without* assuming the ride completed.
- **Completion-Independent Columns:** Features like Avg VTAT and Avg CTAT can be imputed using the median/mean of the respective vehicle type, as they reflect average operational metrics.

B3. Engineer Time and Location Features

- **Temporal Features:** Convert Date and Time into a single timestamp. Extract Hour, Day of Week, and **Peak Hour Flags** (e.g., 7-10 AM, 5-8 PM).
- **Geographical Features:** Encode Pickup Location and Drop Location using methods that capture similarity or density, such as **Target Encoding** based on historical completion rates per zone.

B5. Prevent Data Leakage (Handling Post-Booking Data)

All variables known *only after* ride completion (e.g., Driver Ratings, Customer Rating) must be:

1. **Dropped Entirely:** If they are not relevant for the prediction moment.
2. **Used to Create a Flag:** If their missingness itself is predictive (as in B1), use a binary flag to signal that the data was not available at the time of prediction.

B6. Class Imbalance Check

Analyze the distribution of **Completed** vs. **Non-Completed** events. If the non-completion rate is low (e.g., < 10%), the training data is imbalanced.

- **Strategies:** Use **SMOTE** (Synthetic Minority Over-sampling Technique) or, preferably, **Class Weighting** (assigning a higher penalty to misclassifying the minority class, Non-Completed) during model training.

📊 Section C: Exploratory Data Analysis (EDA)

C1. Distribution Insights

- **Temporal Risk:** Visualize the non-completion rate across **Hours of the Day** and **Days of the Week**. This reveals periods of highest operational stress (e.g., cancellation spikes during Friday evening rush).
- **Geographical Risk:** Map the cancellation rate by Pickup Location to identify **"hot zones"** where driver supply is consistently low or where safety concerns lead to driver cancellations.

C2. Relationship Exploration

- **VTAT/CTAT vs. Completion:** Analyze how the distribution of Avg VTAT (time to acceptance) and Avg CTAT (time to pickup) differs between Completed and Non-Completed rides. High **CTAT** likely drives customer cancellation.
- **Vehicle Type vs. Risk:** Compare the cancellation rate across Vehicle Type (e.g., Auto, Sedan, Bike). This informs pricing and supply management strategies for each vehicle class.

C5. Actionable Insights

- **Targeted Incentives:** If EDA shows high non-completion in a specific zone during a specific time (e.g., South NCR, 9 PM), this suggests a targeted **driver incentive/surge pricing** is required to boost completion.
- **Customer Tiering:** High-cancellation customers identified via historical features may warrant **pre-emptive messaging** or **priority driver matching**.

🤖 Section D: Modeling & Evaluation

D1. Define Target Variable

Target: P(Completion}=1 | features). This is a standard **Binary Classification** problem.

D2. Data Split

A **Time-Based Split** is preferred over a random split, as ride-hailing data is time-series-dependent.

- **Strategy:** Use the earliest 80% of the data by Date for Training and the latest $\text{20\%}$ for Validation/Testing. This ensures the model is tested on *future* conditions, simulating real-world deployment.

D3. Compare Models

Train and compare at least three models:

1. **Baseline: Logistic Regression** (Interpretable baseline for feature importance).
2. **Intermediate: Random Forest** (Handles non-linear relationships and interactions well).
3. **Advanced/Recommended: Gradient Boosting Machine (XGBoost/LightGBM)** (Industry standard for tabular data, capable of high accuracy and handling imbalanced data).

D5. Evaluate Models (The Business Cost of Error)

The final evaluation must move beyond F1/AUC to consider the cost of misclassification .

- **False Positive (FP):** Predicting completion when the ride ultimately **cancels**. **Business Cost:** Driver waste time/fuel, customer frustration, potential loss of both driver and customer.
- **False Negative (FN):** Predicting cancellation when the ride ultimately **completes**. **Business Cost:** Platform unnecessarily intervenes (e.g., assigns a penalty or unnecessary priority to the ride), wasting resources.

The model threshold should be tuned to minimize the **total weighted cost of misclassification**.

🔊 Section E: Insights, Interpretation & Recommendations

E1. Error Analysis

It is likely that **False Positives (FP)**—predicting a ride will complete when it cancels—have a **higher monetary and reputation cost** than False Negatives.

- **Recommendation:** Tune the model's classification threshold to achieve higher **Precision** (lowering FP rate) at the expense of slight Recall, prioritizing a reliable prediction signal for driver dispatch.

E2. What-If Simulations

- **What if we reduce the predicted cancellation risk by 10% in low-supply zones?** This simulates the effect of targeted driver incentives. The business impact is an estimated increase in completed trips and an improvement in driver retention (less unproductive driving).
- **What if we immediately implement a dynamic dispatch penalty for customers with a predicted risk score above 0.8?** This estimates the revenue recovery from high-risk users versus the potential loss of customer volume.

E3. Identify High-Risk Segments

High-risk segments include:

- **Customers with historical high cancellation rates.**
- **Bookings during severe weather/peak rush hours** (VTAT}/CTAT} due to traffic).
- **Specific geographic routes** (e.g., routes crossing jurisdiction lines, known traffic bottlenecks).

E4. Executive Summary

The predictive model successfully predicts ride completion with an estimated AUC of X and a minimized FP rate. By leveraging temporal and historical behavioral features, the model provides an early warning signal for high-risk bookings. The key recommendation is to use the prediction score to trigger **proactive operational interventions**—like dynamic surge pricing and priority driver matching—to stabilize supply and demand in high-risk zones, ultimately reducing driver churn and improving service reliability in the NCR market.

E5. Ethical / Operational Risks

- **Operational Risk (API Latency):** The prediction must be generated in real-time (sub-second) for the dispatch system. **Mitigation:** Use highly efficient models like **LightGBM** and deploy the model in an optimized, low-latency environment.
- **Ethical Risk (Fairness and Bias):** The model might disproportionately penalize customers or drivers in low-socioeconomic areas if geographical zone encoding is used carelessly. **Mitigation:** Monitor model performance and FP/FN rates stratified by location to ensure prediction fairness.